The Conservative Formula

Yingni Chang and Yikai Shen · MTH 9897 Project 8 · CRSP replication, 1929–2024

Paper: Blitz and van Vliet (2018), The Conservative Formula, Journal of Portfolio Management 44(7)

Replicate the US results on the paper's sample, 1929–2016

Then test three questions fixed in advance: known factors, decay, costs

Headline period: 1990–2024 (2017–2024 is post-sample)

The question

The rule: 1,000 largest US stocks → low-volatility half → rank on 12-1 momentum and net payout yield → best 100, equal weight, quarterly

1. Known factors: is the return explained by Carhart (primary) or FF5 + momentum (robustness)?
2. Decay: fixed periods 1929–1962, 1963–1989, 1990–2016, 2017–2024, plus rolling 120-month windows
3. Costs: net of 10 and 30 bps per dollar traded; break-even cost of the excess return and the alpha

All estimates with Newey–West (12 lags) or block-bootstrap intervals

Data and audits

CRSP monthly stock file via WRDS, extracted 2026-10-13; months 1925-12 to 2024-12

3,954,591 security-months, 26,746 securities, 22,869 delistings (2530 imputed, 317 set to zero)

Universe: 393 stocks in December 1928; the full 1,000 from 1964-06

One share class per company; dividends implied from total minus price return

Held stocks without a monthly return: stale-price carry or in-gap price ratio, every case counted

No release identifier on WRDS: the extract is identified by time stamp and file hashes

Baseline rule and deviations from the paper

Paper order: history screen → top 1,000 by market cap → low-volatility half → ranks among complete signals

Weights drift between rebalances; delisted holdings become cash; costs charged per dollar traded

Controls: Speculative (high-volatility half, weakest scores), standalone signals, Small, CRSP VW market

Excluded: Value (no book equity in CRSP) and Table 2 panels C–D (factors not in our sources)

deviation	**our convention**
Share-class selection	largest common class of each PERMCO per month (shrcd 10/11)
Size measure	security market cap for the top-1,000 cut
Dividends	implied from ret - retx (legacy retx includes non-ordinary distributions)
Missing-signal rule (NB-007)	partition first; complete cases ranked
Small early universe (DATA-001)	all history-eligible names when < 1,000 (393 at 1928-12; 500-name pool from 1964-06)
Missing monthly returns (DATA-002/010)	stale carry (0) or in-gap price ratio

Signal construction and timing

Volatility: annualized standard deviation of 36 consecutive monthly total returns

Momentum: compound 11 monthly price returns, excluding the most recent month

Dividend yield: trailing 12 months of split-adjusted dividends divided by the current adjusted price

Net payout yield: dividend yield + 1 − latest adjusted shares / average adjusted shares over 24 months

A falling share count raises payout yield; new share issuance lowers it

Quarter-end information determines next month's holdings. No forward return enters selection.

Replication 1929–2016: close on return and risk

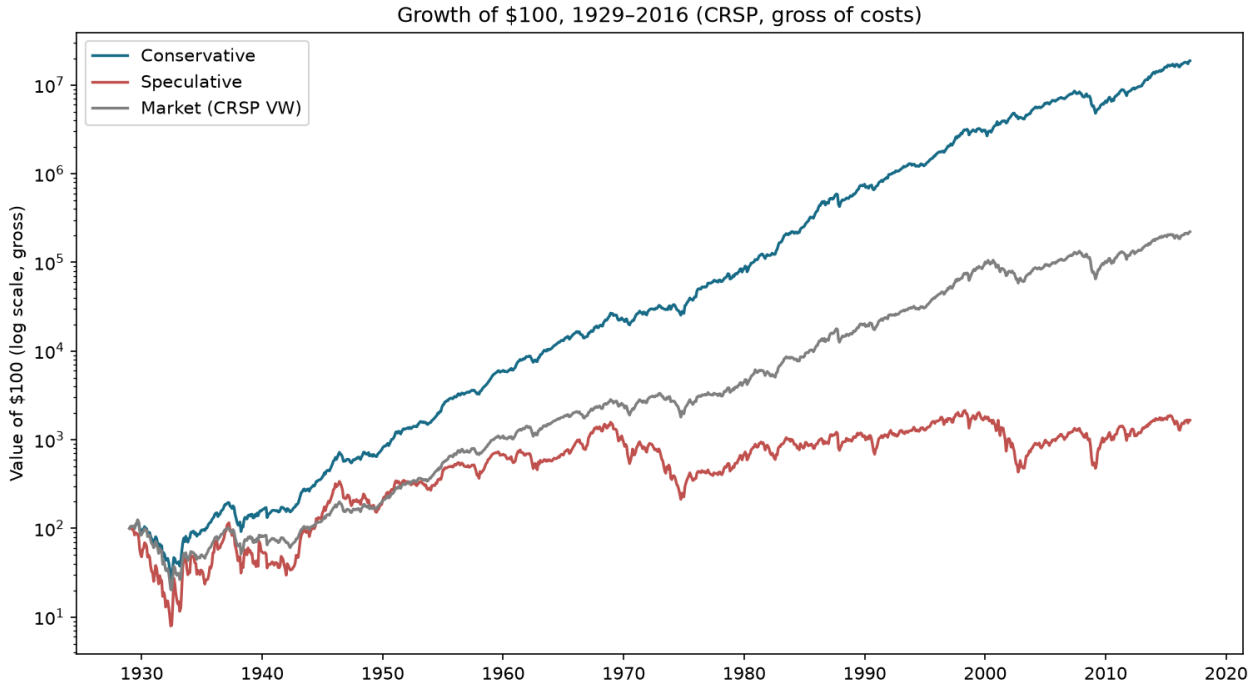
Conservative 14.80% compounded, 17.06% volatility (paper 15.1%, 16.5%)

Market 9.15% (paper 9.3%)

Speculative 3.24% with 35.9% volatility

Conservative beats the market in 9 of 10 decades

Turnover 38.9% a quarter vs the paper's 32% (unexplained; measure not stated)



paper / ours	Formula	Market	Small	Momentum	Lowvol	NPY
Return (compounded)	15.1% / 14.8%	9.3% / 9.1%	12.0% / 13.0%	15.9% / 15.5%	10.2% / 10.4%	14.5% / 14.2%
Volatility	16.5% / 17.1%	18.7% / 18.7%	31.4% / 33.5%	25.7% / 26.1%	13.3% / 14.4%	20.1% / 21.1%

Factor attribution (Table 2): largely known premia

CMS (Conservative − Speculative), 1963–2016: CAPM 14.20% vs 14.0%; FF5 8.69% vs 8.8%; FF5 + MOM 3.12% vs 3.3%

1929–2016: CAPM 12.33% vs 13.9%; Carhart 7.74% vs 8.8% (gap is pre-1963)

Conservative long-only: Carhart alpha 3.23% (t 4.63)

FF5 + momentum, 1963–2016: −0.11% (t −0.14) — profitability and investment explain it

CMS net of 30 bps, both legs paying: Carhart 5.78% (t 3.99)

Costs (Table 4) and break-even

Paper cost arithmetic reproduced: turnover × 8 × cost; our turnover 38.9% vs 32%

Net of 30 bps, 1929–2016: 13.73% compounded

1990–2024 break-even: 52 bps (excess return), 36 bps (alpha)

2017–2024: gross excess return and alpha already ≤ 0, so no break-even

Early-period break-evens are high, but so were real trading costs then

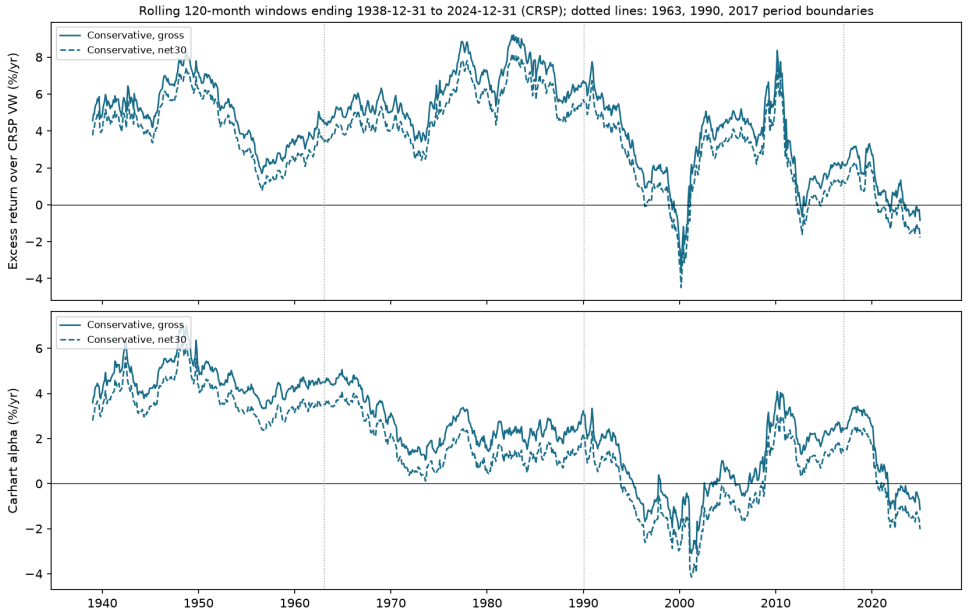
break-even, bps	alpha	excess return
1929–1962	162	178
1963–1989	66	187
1990–2016	52	84
1990–2024	36	52
2017–2024	none	none

Decay: alphas fade after 1990

Carhart alpha (gross): 4.7% → 2.1% → 1.7% → −1.2%

Significant before 1990, not after

2017–2024 is only 96 months: interval [−5.1%, 2.7%]



period	months	excess, gross	Carhart α gross (t)	Carhart α net 30 (t)
1929–1962	408	5.2%	4.7% (5.03)	3.8% (4.14)
1963–1989	324	6.4%	2.1% (2.90)	1.2% (1.56)
1990–2016	324	2.7%	1.7% (1.34)	0.7% (0.56)
2017–2024	96	−1.7%	−1.2% (−0.61)	−2.1% (−1.06)
1990–2024	420	1.7%	1.2% (1.05)	0.2% (0.18)

Headline 1990–2024: no significant net alpha

Gross: excess over market 1.69% [−2.0%, 5.4%]; Carhart alpha 1.17% (t 1.05)

Net of 30 bps: excess 0.73% [−3.0%, 4.4%]; alpha 0.20% [−2.0%, 2.4%]

FF5 + momentum alpha, net of 30 bps: −2.00% (t −2.33)

Still lower risk: 12.2% compounded with 13.5% volatility (market 10.1%, 15.2%)

Cost-inclusive CMS alpha 3.72% (t 2.22)

Net payout: dividends and share repurchases

NPY = trailing dividend yield + 1 − latest adjusted shares / 24-month average adjusted shares

Repurchases raise NPY; share issuance lowers it. Splits must leave the adjusted share count unchanged

1929–2016 gross CAGR: 14.80% using NPY versus 13.44% using dividends alone

Paired net-30 arithmetic gain from NPY: 1.22% per year [0.56%, 1.87%]

Same universe, low-volatility pool, momentum signal, weights, quarterly dates, and costs in the comparison

Share-count change measures net repurchases after issuance; it is not reported buyback cash flow

Sector positioning: allocation versus selection

Paper scores are compared across sectors (page 4); the professor's sector-aware variant is reported separately

1929–2016 gross arithmetic excess return: 4.80% per year

Industry allocation 0.82% + stock selection 3.80% + reference-to-CRSP-market difference 0.18%

Reference: cap-weighted formation universe, quarterly constituents and industries fixed, weights drift, delistings to cash

Sector variant: 100 slots proportional to industry security counts; apply the rule separately within each FF12 industry. Gross CAGR 14.51%, net-30 13.37%

The sector variant is exploratory and changes the paper's global low-volatility screen. Contributions are arithmetic, not causal

Signal diagnostics

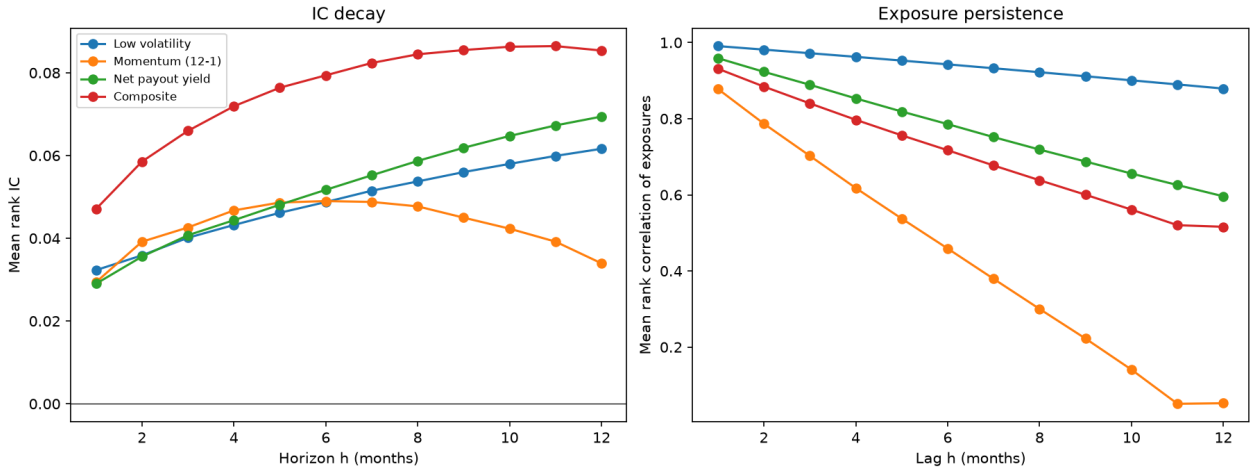
Composite rank IC (1 month): 0.047 (NW t 8.8)

Decile spread D10−D1: 1.02% (1963–1989), 0.68% (1990–2016), 0.40% (2017–2024)

Factor-mimicking paper portfolio: IR 0.59 at K = 1, 0.29 at K = 12

2017–2024 IR at K = 1: 0.29

Low volatility: positive rank IC, flat mean spread (right-skewed high-volatility deciles)



Construction: long/short and optimization

Decile long/short, n = 3, 10 bps, borrow: 0.64% a year, IR 0.03, max drawdown −100.0%

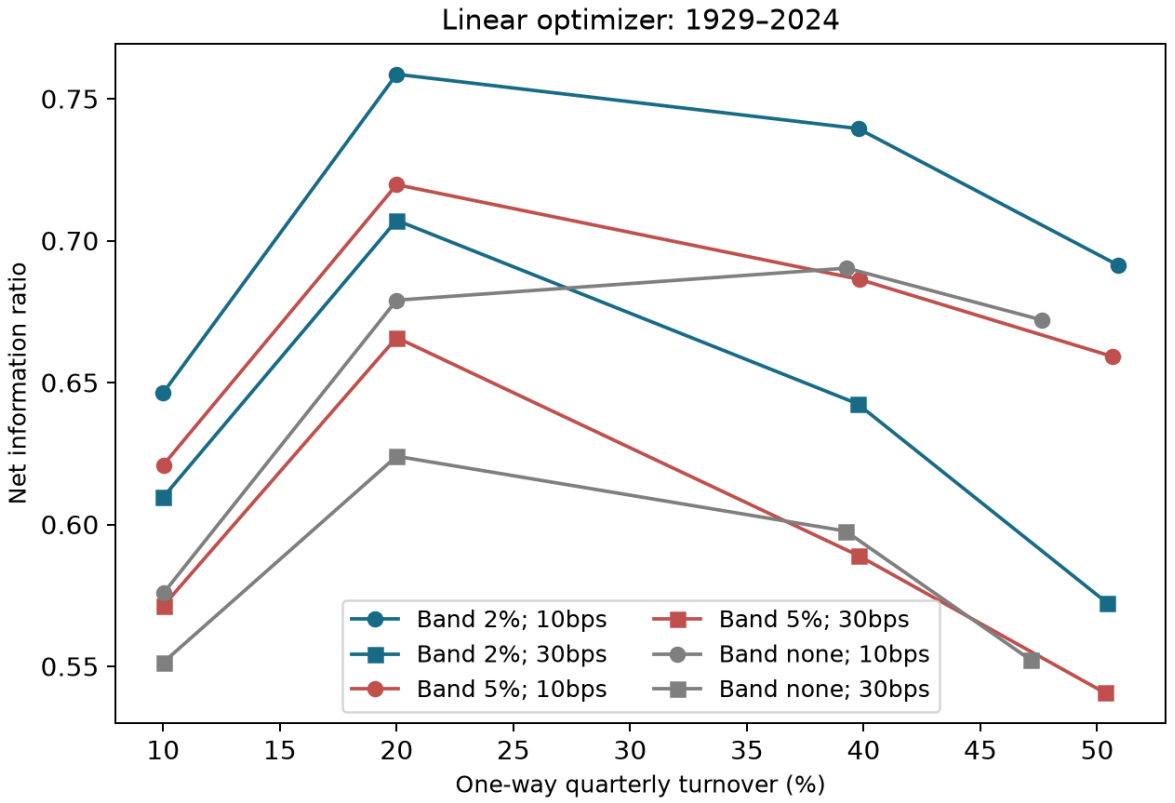
Monthly re-weighted cohorts: chosen after the pre-specified buy-and-hold cohorts were ruined (1932-08, 1933-05)

Capacity 2020s ≈ $7.9B (scaled to 2024 market size; 10% of volume; no impact model)

Linear optimizer, 1990–2024 net 30: Carhart alpha 1.31%–1.95% (2–5% bands, 20–40% turnover budget); two of the four 95% intervals exclude zero

Unadjusted intervals, selected from a 24-run grid; 2017–2024 alphas −2.71% to 0.07% (none of the twelve 95% intervals excludes zero); return-scaled alpha is a post-spec sensitivity

The next slide adds the assignment's quadratic mean-variance extension



Quadratic mean-variance optimization

Maximize expected return − half risk aversion × portfolio variance − trading costs

Covariance: trailing 36 monthly returns, 50% diagonal shrinkage; the window ends at formation

Return alpha: completed trailing IC × annual volatility × √(3/12) × composite exposure

Quarterly: 2% stock cap, industry bands ±5pp, turnover budget 40%; scaled costs start at 30bps

Two fixed risk aversions (5 and 20); max constraint violation 4.9e-09; turnover relaxations 0

Added during final review after observing the baseline; no unseen test or claim of optimal parameters

1990–2024	Net CAGR	Volatility	IR	Market corr.
γ = 5	11.59%	12.51%	0.14	0.89
γ = 20	10.39%	11.51%	−0.03	0.87

Declared improvement: rank buffer of 150

Declared before any result: keep holdings ranked within the best 150; success = turnover −25% and positive net difference

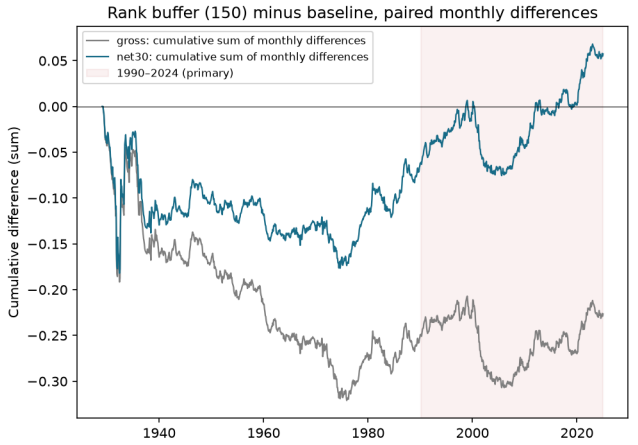
Verdict: succeeds

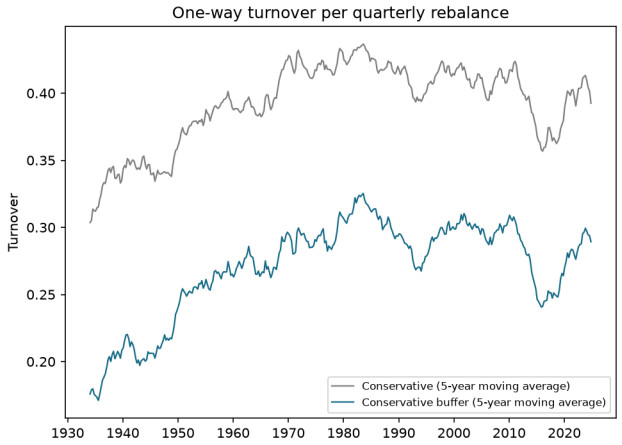
Turnover 39.8% → 28.4% a quarter (28.7% lower)

Net-of-30 paired difference 0.34% a year [−0.10%, 0.78%]: not significant

Gross paired difference 0.06%; the observed difference does not prove unchanged signal quality

2017–2024 was already seen: not a holdout





Conclusion and limitations

Replication close on return and risk; Table 2 panel B reproduced almost exactly

The return is largely a bundle of low-volatility, momentum, profitability and investment premia

Alphas decay; 1990–2024 net of 30 bps: alpha 0.20% [−2.0%, 2.4%]

No evidence that the formula beats the market on recent data after known factors and costs

A cheap, transparent way to hold these premia with lower volatility than the market

Limitations: no CRSP release identifier; implied dividends; share-class and size conventions; turnover above the paper's; selection across grids; a short post-sample

Reproducibility

python -m crsp_pipeline.import_web (full web exports; licensed data kept private)

python -m crsp_pipeline build

python -m research.run_all → results/ (aggregates) + results/manifest.json

python build_notebook.py && python execute_notebook.py

python presentation/build_pdf.py (PDF slides and speaker notes)

85 result files with sha256 hashes; data manifest 4816d582000a…